# Global Earthquake Analysis Using SQL & Streamlit

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**TOOLS:** MySQL, Python, Pandas, Streamlit, Plotly

**DATA SOURCE:** USGS Earthquake Dataset

## PROJECT OVERVIEW

This project performs an end-to-end analysis of global earthquake activity. It includes:

- Loading data into MySQL
- 30 analytical SQL tasks
- A full Streamlit dashboard
- Visualizations and metrics
- Advanced data processing (rms, gap, tsunami alerts, depth categories, etc.)

**Purpose:**

To understand earthquake patterns across magnitude, depth, time, location, continents, casualties, and seismic quality metrics.

## 1. INTRODUCTION

This project focuses on analyzing global earthquake data collected over the last five years and transforming it into meaningful insights using a combination of data preprocessing, SQL, and dashboard development techniques. The workflow includes dataset extraction from a remote source, preprocessing using Python and Pandas, database creation in MySQL, execution of 30 analytical SQL tasks, and the development of an interactive Streamlit dashboard to visualize key patterns, statistics, and trends.

The project demonstrates strong capabilities in data preprocessing, relational database management, analytical querying, and frontend visualization through modern Python-based tools.

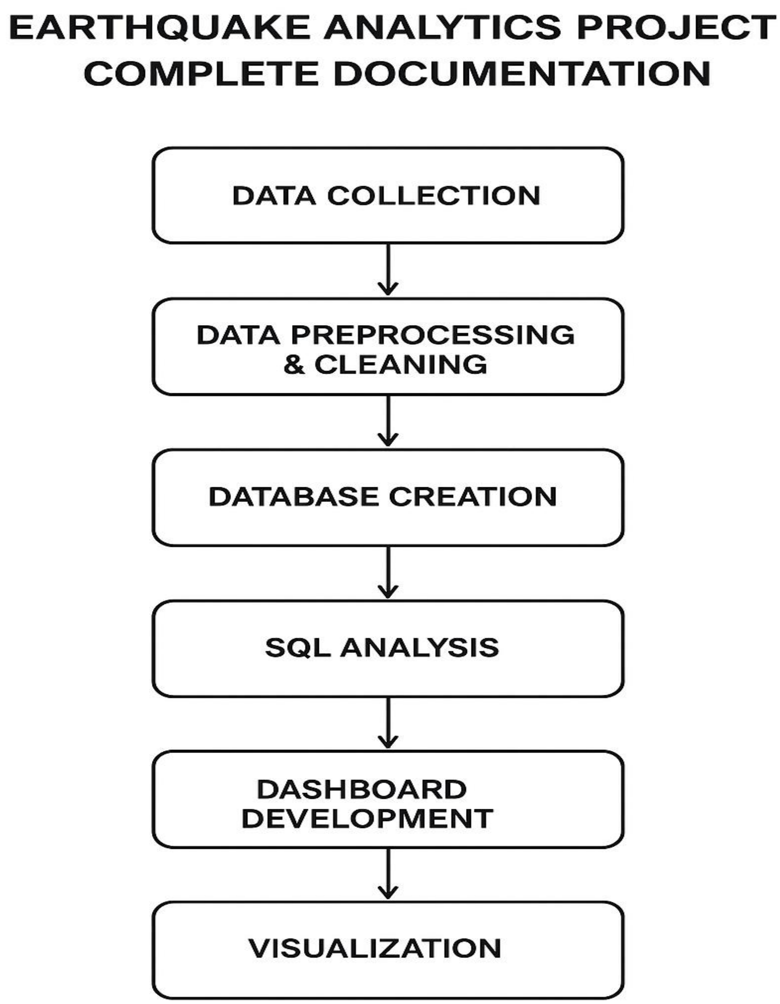


Fig no 1: Earthquake Analytics Workflow

## 2. DATA COLLECTION

The raw earthquake dataset was obtained through a live URL containing global seismic event information from the last five years. The dataset included various attributes such as:

- Event identification details
- Geographic coordinates

- Depth and magnitude
- Location descriptors
- Time-based attributes
- Alert levels, tsunami indicators
- Network information
- Additional metadata such as RMS, GAP, station count, and more

This initial dataset was used as the foundation for further analysis.

## 3. DATA PREPROCESSING & CLEANING

Data preprocessing was performed extensively using Python and the Pandas library to ensure data consistency, integrity, and usability.

A. Handling Missing Values

Several fields included missing values due to incomplete seismic reporting. These fields were processed by:

- Replacing null values with 0/median where appropriate
- Dropping unusable entries
- Standardizing empty or inconsistent records

B. Converting Data Types

Columns such as dates, coordinates, depth, and magnitude were converted to appropriate numerical and datetime types to ensure analytical operations could be performed accurately.

C. Extracting Derived Fields

Additional fields were generated to support analytical queries:

- Year, month, day, hour
- Day of week
- Depth category (shallow, intermediate, deep)
- Country, continent derived from "place" column

These engineered features enabled deeper analytical insights.

D. Cleaning Complex Columns

Some columns contained complex or inconsistent values. For example:

- The properties_types field included comma-separated items and irregular formatting.
- These values were transformed into clean, standardized labels suitable for aggregation and analysis.

E. Final Dataset Creation

After cleaning and preprocessing:

- A final structured dataset was produced
- All columns were standardized
- The dataset was verified for duplicates, inconsistencies, and missing values

This cleaned dataset was ready for integration into a relational database.

## 4. DATABASE CREATION & SQL INTEGRATION

A new MySQL database named earthquake_db was designed to store the cleaned earthquake dataset.

A. Database Structure

- Single table named earthquake
- Contains all relevant attributes from the cleaned dataset
- Proper data types assigned (INT, FLOAT, VARCHAR, DATETIME, etc.)

B. Data Loading

- The cleaned dataset was inserted into the database in a structured format.

C. SQL Analytical Tasks

A total of 30 SQL tasks were executed to extract meaningful insights. These tasks covered:

- Ranking-based queries (top depth, top magnitude)
- Time series analysis (year, month, day trends)
- Magnitude and depth relationships

- Tsunami occurrences
- Continent-wise and country-wise statistics
- Alert-level performance
- Reliability score assessment
- Advanced queries using joins, window functions, and conditional aggregation
- Geographical pattern detection
- Maximum impact events
- Shallow-to-deep ratio analysis
- High-frequency seismicity regions

These SQL tasks comprehensively explored the dataset from multiple analytical perspectives.

## 5. STREAMLIT DASHBOARD DEVELOPMENT

An interactive Streamlit dashboard was developed to visualize earthquake statistics and support user-driven data exploration.

A. Multi-Page Application Structure

The dashboard includes two dedicated sections:

1. Home Dashboard – Shows filters, KPIs, and visualizations
2. SQL Tasks Page – Allows users to view the results of all 30 SQL tasks

B. Filters

The Home Dashboard includes filters that allow users to dynamically adjust the dataset view:

- Year filter
- Country filter

These filters affect the dataset preview, KPIs, and charts.

C. Data Preview Section

- Users can view the filtered dataset in a tabular format.

- This helps validate entries and understand the raw seismic information.

D. Key Performance Indicators (KPIs)

The dashboard displays the following metrics:

- Total number of earthquakes
- Average magnitude
- Number of tsunami-related events
- Most active region

Each KPI updates dynamically based on selected filters.

E. Visual Analytics

Multiple interactive charts were developed using Plotly:

- Magnitude distribution
- Earthquakes per year
- Depth category distribution
- Top 20 earthquake-active countries

These visualizations allow the user to identify major patterns, outliers, and trends.

F. SQL Query Section

- A user-friendly SQL editor was added where users can input custom SQL queries and immediately view the output.
- This makes the dashboard highly interactive and suitable for analytics exploration.

G. Task Page Navigation

The dashboard includes navigation that allows users to switch from:

- Home → Tasks page
- Tasks page → Home

The Tasks page contains all 30 pre-defined SQL analysis outputs.

## 6. KEY ANALYTICAL INSIGHTS

Through the 30 SQL tasks and visual analysis, the project reveals several important insights:

- Years with the highest seismic activity
- Months and hours where earthquakes occur most frequently
- Countries with significant shallow or deep earthquake profiles
- Regions experiencing severe high-magnitude events
- Alert-level patterns and tsunami correlations
- Network performance and reliability indicators
- Geographical distribution of high-risk zones
- Economic loss patterns
- Magnitude trends across regions and continents
- Year-over-year seismic activity growth

These insights provide actionable intelligence for research, disaster preparedness, and environmental studies.

## 7. FUTURE ENHANCEMENTS

To make the project more robust and enhance its capabilities, several improvements can be added:

A. Machine Learning Integration

Predictive models can be built for:

- Magnitude forecasting
- Tsunami probability
- Depth classification
- Regional risk assessment

B. Real-Time API Integration

USGS or other seismic data APIs can be connected to update the dashboard in real-time.

C. Advanced Geospatial Maps

Tools such as Folium or Kepler.gl can be used for:

- Heatmaps
- Cluster maps
- Interactive world maps

D. Automated Reporting

Weekly or monthly PDF reports with charts and KPIs can be generated automatically.

## 8. CONCLUSION

This project demonstrates a complete end-to-end data analytics workflow, starting from raw data ingestion to dashboard deployment. Key strengths include:

- Data extraction from live sources
- Advanced preprocessing and feature engineering
- Clean normalization and database integration
- Execution of 30 analytical SQL tasks
- Development of a fully functional interactive Streamlit dashboard
- Clear visualization of trends, relationships, and earthquake behavior

This work showcases strong capabilities in:

- SQL
- Python
- Pandas
- Data visualization
- Application development
- Analytical storytelling

The final dashboard provides both technical and non-technical users with an intuitive and insightful interface to explore global earthquake activity.